

New Threats = Increase in operational complexity and manageability?

The current security defense paradigm is to deploy more and more of the existing security technologies throughout every segment of the network. This includes firewalls and ACLs to block access and perform application inspection, intrusion protection system (IPS) technology to provide very granular traffic inspection and identify known threats, encryption software to counter eavesdropping, anomaly detection to detect worms or DoS attacks, and antivirus software to battle viruses.

Many of today's security technologies were developed to perform their specific function with little context of the overall network threat environment. Operating alone, however, these technologies are less effective in stopping the newer attacks, as well as the changing ways in which user's access networks, because of the "security gaps" that exist between each technique's capabilities. With the increased complexity of threats, such as the blended threats that use a combination of techniques to disrupt networks, security technologies must operate in a coordinated fashion to stop attacks and better control network activity and applications.

Unfortunately, over the years, many companies have addressed nagging security concerns by constantly adding devices and software to address each particular problem. This has led to separate antivirus protection, firewalls, VPNs, and intrusion prevention. While this addresses the short-term needs, it creates an entirely new and bigger problem: managing multiple systems that operate independently of one another. As more advanced threats emerge, there is a need for network security to become more holistic; security technologies must act in coordination to detect and defend against more sophisticated threats. There is a growing need for devices that can assemble the puzzle pieces and lock down the gaps that exist in conventional network security systems.

Adaptive security for a changing world

Transforming chaos into clear and manageable security policy is essential, which is why future network security systems need to focus on convergence and consolidation. For robust information security for an enterprise, a proactive architectural and system approach is critical. The idea is to accurately identify and stop attacks as early and as far from the destination host as possible, while simultaneously simplifying the security architectures required to do this.

Instead of a security product for each security need in isolation, an end-to-end security solution approach or system enables these

combined functions to operate as a coordinated defence (instead of silos) that stops a broader range of attacks and greatly reduces the number of diverse multi-vendor devices that must be deployed, thereby simplifying security design and management.

Historically, firewalls have generally been considered fairly simple devices, but they are effective at what they do: either block a packet or let it through based on Layer three and Layer four information and session state. They can provide some level of application inspection but do not perform the detailed inspection of some other technologies. An IPS device can pick up where a traditional firewall leaves off by peering more deeply into a packet's contents to see whether the data within conforms to company policy. Host IPS, host firewall on servers and desktops/laptops, day zero protection and intelligent behavioral based protection from application vulnerability and related flaws (within or inserted by virus, worms or Trojans) provided great level of confidence on what is happening within an organisation on a normal day and when there is an attack situation, which segment and what has gone wrong and gives flexibility and control to stop such situations by having linkages of such devices with monitoring, log-analysis and event co-relation system.

Adaptive threat defense is the need now where an end-point security system can dynamically generate an attack signature and push it to the other end-points and to the perimeter IPS devices to stop it from propagating to rest of the infrastructure. Similarly security operations teams have challenges to fine-tune false positives, they struggle by logging into various devices to understand which logs are resulting into what logs; the need of the hour is that security operations team should be able to easily see the logs and link the same to configurations which are resulting into these logs.

This type of systems approach transforms security from operating as separate siloed technologies in a reactive mode – with limited and static detection methods – to functioning as a coordinated, proactive threat defense system that adapts to the threat environment.

These systems provide numerous benefits: improved detection, greater event classification accuracy, lower operating costs, streamlined administration, and services extensibility that integrate the most advanced security technologies as they are developed. Most importantly, these converged systems will not compromise the quality of security in any given category, but instead combine the strength of each in complementary ways to deliver a tighter, coordinated defence. ■

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